

# Consistency by Construction: Cross-Encoders with Reversal-Aware Decoding for SemEval-2027 Task 2 (DiCo-NLI)

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**Abstract**—DiCo-NLI asks a system to label an ordered phrase pair as equivalence, forward entailment, backward entailment or another relation, and then checks whether the labels it gives to  $(a, b)$  and  $(b, a)$  agree. We fine-tune multilingual cross-encoders and move consistency out of training and into decoding: the log-probabilities of a pair and its reversed twin are mapped into one frame, averaged, and decoded once, so every pair predicted as reversible is consistent by construction. Almost all remaining errors lie in deciding whether a pair is in an entailment relation at all. Six methods aimed at that decision, including focal loss, extra iSTS and WordNet data and a stacked classifier, did not beat a plain ensemble. What did help is a hierarchical composition: ordinary four-way models decide the negative class, and models trained only on the three entailment labels choose among them. Averaged over the English, Spanish, Basque and mixed tracks of the development set, this raises Weighted F1 / SoftCons / HardCons from 80.8 / 95.7 / 88.0 to 81.4 / 96.8 / 89.5 with fixed members, and to 83.7 / 98.6 / 91.7 with members selected per track. If the reversible/negative split is read from the layout of the input files, a selected ensemble reaches 94.3 / 100 / 93.1, above the current development leader (91.4 / 100 / 89.7). We report that variant separately because it depends on how the data were packaged.

**Index Terms**—natural language inference, directional consistency, cross-encoders, multilingual NLP, Basque, SemEval

## I. INTRODUCTION

Natural language inference (NLI) models are usually scored one example at a time. A model can do well on that score and still contradict itself, for example by labelling both  $(a, b)$  and  $(b, a)$  as forward entailment, which would make each phrase strictly more specific than the other. SemEval-2027 Task 2, DiCo-NLI [1], [2], makes this visible. Every reversible phrase pair appears in both orders, and systems are scored on per-example Weighted F1 and on two consistency measures over the pairs.

Our system is built around one observation: consistency can be enforced at decoding time for free, because both directions of every reversible pair are in the same input file. We average the two directions’ scores in a shared frame and choose a single label for the pair. This leaves the classifier with one job, which is to separate the entailment family from everything else. That job turned out to be the hard part. This report describes the decoding scheme and two exactly reversal-equivariant model variants; compares seven backbones, including two multilingual rerankers; profiles where the errors come from; and reports a negative result, since none of the six NEG-focused methods we tried improved on a plain ensemble.

The method that did help reuses models trained without the negative class inside a hierarchical composition, and per-track ensemble selection adds more on the development set. Code, configurations and run logs are public.<sup>1</sup>

## II. TASK AND DATA

Each instance is an ordered pair  $(a, b)$  of short phrases, about six words in total, taken from the chunk alignments of PhrasIS [3], which in turn come from the interpretable STS datasets of SemEval 2015–2016 [4]–[6]. The label is EQ, FE ( $a \models b$ ), BE ( $b \models a$ ) or NEG, where NEG covers every other PhrasIS relation (similar, related, opposite). The reversal map is  $\text{Rev}(\text{EQ}) = \text{EQ}$ ,  $\text{Rev}(\text{FE}) = \text{BE}$  and  $\text{Rev}(\text{BE}) = \text{FE}$ ; NEG has no reverse. There are four tracks: English (T1), Spanish (T2), Basque (T3) and mixed-language pairs (T4).

Weighted F1 is the support-weighted F1 over the four labels. SoftCons counts a gold-reversible pair as consistent when both predictions are reversible and Rev-compatible, whatever the gold label. HardCons requires both directions to be correct. Predicting NEG on either side of a gold-reversible pair therefore costs both consistency scores at once, which makes a false NEG the most expensive error.

We checked three facts on every training and development file. Instance ids have the form `<pair_id>__<l1>-<l2>__<original|flipped>`, and the reversed twin swaps both the language codes and the direction tag. Every reversible instance has its twin in the same file, while every NEG instance appears once. A `pair_id` is shared across tracks, with two rows in each monolingual track and twelve in T4. Each monolingual track has 3,042 training rows (1,282 reversible pairs and 478 NEG) and 660 development rows (277 and 106).

Table I splits the English pairs by how their words overlap, which turns out to predict where models go wrong. When the words of one phrase are contained in the other (*in a field/in a huge field*), the pair is almost always directional: 418 of 458 training pairs are FE or BE and only 13 are NEG. When the phrases share no word at all, EQ (142) and NEG (114) dominate, and telling them apart needs lexical knowledge: *climbs/moves up* is EQ, while *along a runway/on the tarmac* is NEG. Most NEG pairs (351 of 478) fall in the middle group, where the phrases share some words and differ in

<sup>1</sup><https://github.com/MinLee0210/dico-nli-challenge>

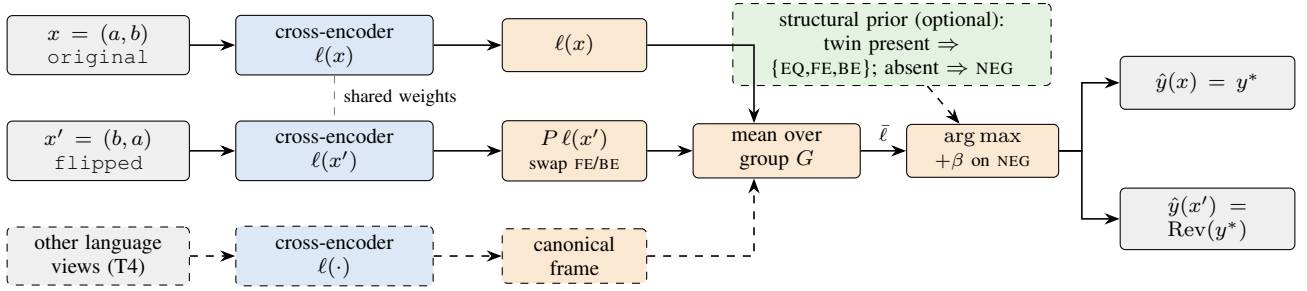


Fig. 1. Consistency-aware decoding. Scores of an instance and its reversed twin (and, in T4, of other language views of the same source pair) are mapped into the original frame, averaged, and decoded once. The reversed row receives the reversed label, so the pair is consistent whenever it is not NEG. The optional structural prior restricts the choice using the file layout.

TABLE I  
ENGLISH (T1) SOURCE PAIRS BY LEXICAL OVERLAP BETWEEN THE TWO PHRASES, WITH THE GOLD LABEL DISTRIBUTION. “CONTAINED”: THE WORD SET OF ONE PHRASE IS A STRICT SUBSET OF THE OTHER.

| Split | Overlap   | EQ  | FE  | BE  | NEG        | Total |
|-------|-----------|-----|-----|-----|------------|-------|
| train | none      | 142 | 55  | 54  | 114        | 365   |
|       | partial   | 230 | 163 | 191 | <b>351</b> | 935   |
|       | contained | 27  | 218 | 200 | 13         | 458   |
| dev   | none      | 27  | 8   | 8   | 27         | 70    |
|       | partial   | 53  | 39  | 52  | <b>75</b>  | 219   |
|       | contained | 7   | 47  | 36  | 4          | 94    |

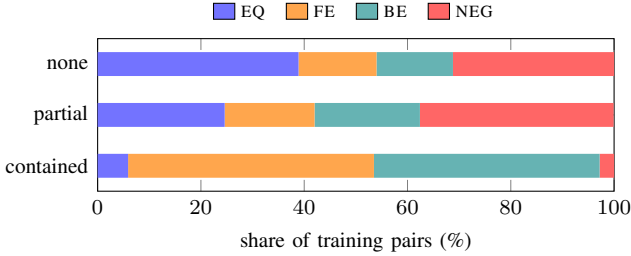


Fig. 2. Label mix of the English training pairs in each overlap group of Table I.

others (*on Monday/on Thursday, across the sand/through the water*). Only 43 training pairs contain two different numbers, so numeric mismatch is a minor error source. Fig. 2 shows the label mix of each group.

### III. RELATED WORK

Interpretable STS [5], [6] introduced the chunk-level relation types from which PhrasIS and DiCo-NLI are derived. Consistency of neural models under logical transformations has been studied for NLI [7], [8], usually through extra loss terms. For symmetric pair tasks, Xu et al. [9] and Kumar and Joshi [10] penalise disagreement between the two input orders; we use the same idea for an antisymmetric label set, where the two orders should agree up to the Rev map. Our NEG study draws on work showing that NLI models lean on lexical overlap [11] and fail on simple lexical inferences [12], and on focal loss [13] and intermediate-task transfer [14] as remedies. For Basque we rely on multilingual encoders; XLM-R-large

was the strongest of the multilingual encoders evaluated on XNLIeu [15]. Lexical knowledge in NLI has often come from WordNet [16]; the Open Multilingual Wordnet [17] and the Multilingual Central Repository [18] link Spanish and Basque wordnets to the same synsets, which we use to build training pairs for all three languages. Beyond NLI checkpoints, we try two multilingual rerankers of a different pretraining lineage, BGE-M3’s reranker [19] and the Qwen3 reranker [20]. Our ensembles use greedy forward selection from a library of models [21].

## IV. SYSTEM

### A. Cross-encoder

A pretrained encoder reads  $[\text{CLS}] a [\text{SEP}] b [\text{SEP}]$  and a linear head maps the first-token state to four logits [22], [23]. We use DeBERTa-v3 [24], [25] for English only, and three multilingual backbones: mMBERT-base [26], an mDeBERTa-v3-base checkpoint fine-tuned on multilingual NLI data [27], and XLM-R-large [28] fine-tuned on XNLI [29], [30]. Multilingual models are trained on all four tracks at once (27,378 rows), so each source pair is seen in nine language combinations.

### B. Consistency-aware decoding

Fig. 1 shows the decoding pipeline. For an instance  $x$  with log-probabilities  $\ell(x) \in \mathbb{R}^4$ , the canonical score is  $\tilde{\ell}(x) = \ell(x)$  for original rows and  $\tilde{\ell}(x) = P \ell(x)$  for flipped rows, where the permutation  $P$  swaps the FE and BE coordinates. Instances are grouped and the mean of  $\tilde{\ell}$  over a group is decoded to one canonical label, after adding a bias  $\beta$  to the NEG score:

$$y^* = \arg \max_c \left( \frac{1}{|G|} \sum_{x \in G} \tilde{\ell}_c(x) + \beta [c = \text{NEG}] \right). \quad (1)$$

Original rows receive  $y^*$  and flipped rows receive  $\text{Rev}(y^*)$ . With *independent* grouping each instance is its own group, which is plain argmax. *Twin* grouping joins an instance with its reversed twin. *Source* grouping joins every row with the same `pair_id` in the decoded file, which pools language views in T4. Any group holding both directions is Rev-consistent by construction. All per-track results in this report decode each track file on its own.

### C. Symmetric architecture

Decoding gives consistency only when both directions are in the input. We also built a model that is consistent for any single input. It scores both orders with the same encoder  $f$  and combines them as

$$g(a, b) = \frac{1}{2}(f(a, b) + P f(b, a)), \quad (2)$$

so  $g(b, a) = P g(a, b)$  holds exactly (Fig. 4). The cost is two forward passes per example. A unit test checks the identity to within  $10^{-5}$ .

### D. Entailment-factorized head

The flat head throws away the 3-way NLI head of the checkpoints we start from. The four DiCo labels are, however, the four combinations of two entailment decisions, so we also built a head that keeps it. With  $q = p(\text{entailment} \mid a, b)$  and  $q' = p(\text{entailment} \mid b, a)$  read from the unchanged NLI head, the label distribution is

$$\begin{aligned} p(\text{EQ}) &= q q', & p(\text{FE}) &= q(1 - q'), \\ p(\text{BE}) &= (1 - q) q', & p(\text{NEG}) &= (1 - q)(1 - q'). \end{aligned} \quad (3)$$

The four terms sum to one, swapping  $a$  and  $b$  swaps  $q$  and  $q'$  and hence FE and BE, so the head is exactly reversal-equivariant like the symmetric model, and it starts from a working entailment classifier instead of a random layer. Training minimises cross-entropy on the composed distribution; the consistency term is zero by construction and is switched off.

### E. Training objectives

The loss is cross-entropy, optionally with label smoothing, plus a reversal-consistency term between each row and its twin, which a twin-aware batch sampler keeps in the same batch:

$$\mathcal{L} = \text{CE} + \lambda \text{KL}_{\text{sym}}(p(x) \parallel P p(x')). \quad (4)$$

For the NEG study we added class-weighted and focal [13] variants of the cross-entropy term and a *reverse-negative* augmentation that adds the reversed copy of each NEG pair; NEG is its own reverse, so the label stays correct.

### F. Structural prior

The second data fact makes the reversible-or-NEG decision readable from the file: a group whose reversed twin is present is decoded among EQ, FE and BE, and a group without a twin is decoded as NEG. This is exact on training and development data. It is a property of how the files were built, not of the phrases, and the test files may be packaged differently, so we report every result with and without it. *Prior-aware* models are trained on the three reversible labels only, since under the prior the model is never asked about NEG; they must not be used without it.

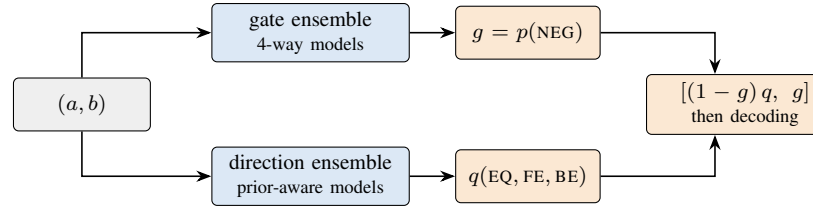


Fig. 3. Hierarchical composition. Ordinary four-way models decide whether the pair is NEG; prior-aware models, trained on the reversible labels only, decide which reversible label it is. No file-layout information is used.

### G. Hierarchical composition

Prior-aware models turn out to be better at choosing among EQ, FE and BE than models that also learn NEG, because none of their capacity goes to the fourth class. We reuse them without the prior by composing two ensembles (Fig. 3). A *gate* ensemble of ordinary four-way models gives  $g = p(\text{NEG})$ , and a *direction* ensemble of prior-aware models gives a distribution  $q$  over the three reversible labels:

$$p(\text{NEG}) = g, \quad p(c) = (1 - g) q(c), \quad c \in \{\text{EQ}, \text{FE}, \text{BE}\}. \quad (5)$$

The composed scores go through the same consistency-aware decoding as any other model. Nothing here reads the file layout: the gate decides NEG from the phrases alone.

### H. Lexical-relation data from WordNet

To give the models the hypernym-versus-sibling knowledge that the error analysis (Section VI-D) points to, we generate short pairs from the Open Multilingual Wordnet [17] through the `wn` library [31]. English, Spanish and Basque synsets share interlingual identifiers, so one English hypernym graph labels all three languages. A hyponym with its direct hypernym gives FE (*creek*  $\models$  *stream*), with the reversed row as BE; two lemmas of one synset give EQ; two siblings under the same hypernym give NEG (*creek/river*), written once as DiCo writes NEG pairs. We draw 600 pairs per label and language, 30% of them with the two phrases in different languages as in T4, for 9,000 rows in total, and add them to the training data. The lemmas come from the whole wordnet, so many are rarer than the caption and headline vocabulary of DiCo.

### I. Ensemble selection

With about forty trained models, we choose ensemble members by greedy forward selection [21]: start from the best single model and repeatedly add, with replacement, the model that most raises the mean of the three official scores, stopping when no addition helps. Each track is ranked on its own, so we select a separate ensemble per track, which also lets T1 use English-only models. For the hierarchical system, each step may add a model to either the gate or the direction ensemble. Selection uses the development labels, so selected scores are optimistic estimates of test performance; we report fixed ensembles alongside them.

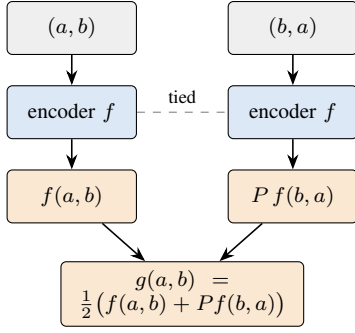


Fig. 4. The symmetric model scores both input orders with one encoder and averages the results after swapping FE and BE on the reversed side. Swapping  $a$  and  $b$  therefore swaps FE and BE in the output exactly.

## V. EXPERIMENTAL SETUP

All models were trained on one NVIDIA RTX 4000 Ada (20GB) with mixed precision, AdamW [32], 10% linear warm-up and linear decay, batch size 16–32 and maximum length 64–128 subword tokens. Learning rates were  $2 \cdot 10^{-5}$  for DeBERTa-v3-base,  $5 \cdot 10^{-5}$  for mmBERT,  $1.5 \cdot 10^{-5}$  for mDeBERTa and  $10^{-5}$  for XLM-R-large. The symmetric XLM-R model used batch size 8 to fit in memory. The multilingual models reach their best score within the first epoch, so for the main runs we validate every 15% of an epoch and stop after five checks without improvement. The factorized-head runs (Section IV-D) used the same settings on one NVIDIA L4 (24 GB), and the reranker, WordNet and later prior-aware runs on one NVIDIA A40 (48 GB), up to four at a time. bge-reranker-v2-m3 uses a learning rate of  $10^{-5}$ ; Qwen3-Reranker-0.6B, a decoder, reads the pair through a short prompt (*Phrase A: ... Phrase B: ... Relation:*) with bf16 autocast and a learning rate of  $2 \cdot 10^{-5}$ , and is classified from its last token. We keep every model at or below about 600M parameters. All runs are logged to Weights & Biases.

The early-stopping monitor is the mean of the three official scores with the four development files decoded together. It is only used to stop training and to compare runs quickly; every table decodes each track separately. Ensembles average log-probabilities before decoding. Our metric code agrees with the official scorer to machine precision on all tracks and decoding modes, and the headline results were re-scored with the official scorer.

## VI. RESULTS

### A. Decoding

Table IV isolates decoding on a single DeBERTa-v3-base model for T1. Twin decoding costs 0.6 F1 points and gains 5.4 SoftCons and 2.5 HardCons points. SoftCons stays below 100 only because some reversible pairs are predicted as NEG. With the structural prior, the same model scores exactly what the development leader reports for T1 (92.4 / 100 / 91.0), which suggests the leader uses the same prior.

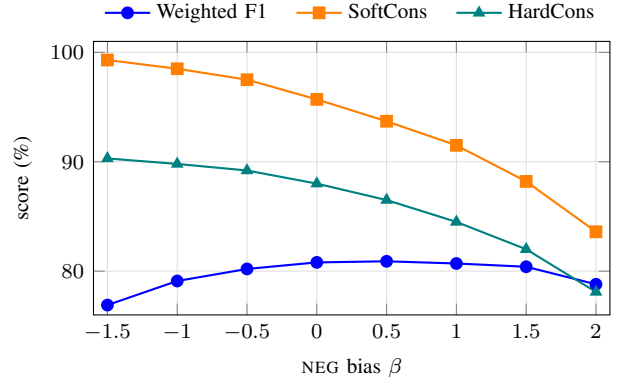


Fig. 5. Effect of the NEG bias  $\beta$  on the clean seven-model ensemble (ALL-4 development scores). Lowering  $\beta$  trades F1 for consistency.

### B. Backbones and model variants

Tables II and III give per-track development scores without and with the structural prior. The two NLI-trained encoders, mDeBERTa and XLM-R-large, are close to each other and well ahead of mmBERT, which starts from masked-language-model weights and overfits after one epoch. A Basque-only encoder trained on T3 alone (JaunBERT) reached a monitor score of only 76.9, so we kept the multilingual models for Basque. The symmetric architecture does not help a single model, and adding it to the clean ensemble lowers HardCons from 88.0 to 87.3. With consistency already enforced at decoding time, the architectural guarantee adds nothing here.

On T1, the English-only DeBERTa-v3-large reaches 83.3 / 95.3 / 88.1, but the multilingual ensemble already scores 83.7 / 95.3 / 89.9 and adding the large model does not change it. Seed and backbone ensembles give the only consistent gains. The best clean system averages seven models (four mDeBERTa and three XLM-R-large seeds) and reaches 80.8 / 95.7 / 88.0 over the four tracks.

With the prior, the prior-aware models are the strongest single models, and the best system averages fourteen of them. It reaches 92.6 / 100 / 91.2 over the four tracks, above the development leader (91.4 / 100 / 89.7) on T1, T3, T4 and the average, and slightly below it on T2.

The two rerankers, added for diversity, are weaker than the NLI checkpoints. bge-reranker-v2-m3 reaches 75.7 / 96.2 / 82.6 without the prior, three to four F1 points below mDeBERTa and XLM-R-large. Qwen3-Reranker-0.6B does worse still (62.6 F1): it almost never predicts NEG, which gives it the highest SoftCons in Table II and the lowest F1. With the prior both stay behind the NLI encoders, so relevance-ranking pretraining does not transfer well to this label set.

The WordNet pairs of Section IV-H move F1 in the right direction for mDeBERTa (80.0 against 79.4) at a cost in consistency (93.4 SoftCons against 95.9), and hurt XLM-R-large on every metric. In the prior-aware models they help neither the plain nor the symmetric variant. We read this as a domain problem: pairs such as *filarialnematoda* teach relations

TABLE II  
DEVELOPMENT RESULTS WITHOUT THE STRUCTURAL PRIOR (F1 = WEIGHTED F1, S = SOFTCONS, H = HARDCONS, IN %). A = mDeBERTa, FINE-GRAINED VALIDATION; B = XLM-R-LARGE XNLI. SINGLE SEEDS EXCEPT ENSEMBLES. BEST VALUE PER COLUMN IN BOLD.

| System                          | T1 EN       |             |             | T2 ES       |              |             | T3 EU       |              |             | T4 Mixed    |              |             | ALL-4       |             |             |
|---------------------------------|-------------|-------------|-------------|-------------|--------------|-------------|-------------|--------------|-------------|-------------|--------------|-------------|-------------|-------------|-------------|
|                                 | F1          | S           | H           | F1          | S            | H           | F1          | S            | H           | F1          | S            | H           | F1          | S           | H           |
| mmBERT-base                     | 77.4        | 87.7        | 78.3        | 74.1        | 91.7         | 78.0        | 67.8        | 88.8         | 73.6        | 72.5        | 83.4         | 74.4        | 72.9        | 87.9        | 76.1        |
| mDeBERTa-v3-base NLI            | 83.6        | 95.3        | 89.2        | 79.3        | 94.2         | 86.6        | 77.4        | 93.5         | 84.1        | 78.9        | 94.2         | 85.9        | 79.8        | 94.3        | 86.5        |
| mDeBERTa, fine-grained val. (A) | 81.8        | 96.0        | 88.4        | 78.2        | 94.9         | 86.3        | 78.8        | 95.3         | 86.3        | 78.8        | 97.5         | 87.0        | 79.4        | 95.9        | 87.0        |
| A + rev.-neg. aug.              | 83.4        | 92.1        | 86.3        | <b>81.1</b> | 92.4         | 84.5        | 76.4        | 89.2         | 79.1        | 80.0        | 91.3         | 82.7        | 80.2        | 91.2        | 83.1        |
| A + NEG weight 2                | 82.8        | 92.4        | 85.9        | 80.5        | 93.9         | 85.9        | 76.7        | 88.4         | 79.4        | 78.7        | 90.3         | 81.2        | 79.7        | 91.2        | 83.1        |
| A + focal ( $\gamma=2$ )        | 81.4        | 94.9        | 88.1        | 78.5        | 94.9         | 86.6        | 75.3        | 94.2         | 84.5        | 77.6        | 93.5         | 84.5        | 78.2        | 94.4        | 85.9        |
| A + iSTS answers-students       | 80.6        | 96.0        | 88.4        | 76.8        | 95.3         | 85.6        | 76.2        | 95.7         | 85.9        | 77.5        | 97.8         | <b>88.1</b> | 77.8        | 96.2        | 87.0        |
| A, symmetric arch.              | 82.1        | 93.1        | 85.6        | 78.6        | 93.5         | 84.1        | 77.2        | 92.4         | 82.7        | 79.4        | 94.2         | 84.8        | 79.3        | 93.3        | 84.3        |
| A, entail2 head                 | 82.6        | 93.9        | 88.8        | 77.2        | 96.0         | 83.8        | 75.4        | 95.7         | 83.0        | 78.5        | 97.8         | 87.4        | 78.4        | 95.8        | 85.7        |
| A + WordNet pairs               | 81.9        | 92.8        | 86.3        | 79.1        | 94.2         | 84.5        | 78.4        | 93.1         | 84.8        | <b>80.6</b> | 93.5         | 86.6        | 80.0        | 93.4        | 85.6        |
| XLM-R-large XNLI (B)            | 83.0        | 95.3        | 88.1        | 79.2        | 96.4         | 87.4        | 75.1        | 96.8         | 82.7        | 77.9        | 96.8         | 87.0        | 78.8        | 96.3        | 86.3        |
| B + rev.-neg. aug.              | 78.5        | 92.1        | 83.0        | 75.1        | 93.9         | 80.9        | 70.1        | 91.0         | 75.5        | 73.3        | 93.9         | 79.1        | 74.3        | 92.7        | 79.6        |
| B + iSTS answers-students       | 80.4        | 93.1        | 85.2        | 76.6        | 92.1         | 83.8        | 75.7        | 93.5         | 83.0        | 77.8        | 92.4         | 85.2        | 77.6        | 92.8        | 84.3        |
| B + WordNet pairs               | 82.0        | 96.4        | 88.1        | 77.6        | 95.7         | 85.2        | 72.8        | 96.4         | 80.9        | 76.9        | 94.6         | 84.8        | 77.3        | 95.8        | 84.7        |
| bge-reranker-v2-m3              | 79.1        | 97.1        | 86.3        | 76.4        | 96.4         | 83.8        | 71.8        | 96.0         | 78.7        | 75.5        | 95.3         | 81.6        | 75.7        | 96.2        | 82.6        |
| Qwen3-Reranker-0.6B             | 68.2        | <b>99.3</b> | 82.7        | 64.3        | <b>100.0</b> | 79.4        | 56.0        | <b>100.0</b> | 70.4        | 61.9        | <b>100.0</b> | 78.3        | 62.6        | <b>99.8</b> | 77.7        |
| Ens. clean-7 (A, B + 5 seeds)   | <b>83.7</b> | 95.3        | <b>89.9</b> | 80.6        | 96.0         | <b>88.1</b> | <b>79.4</b> | 95.7         | <b>87.0</b> | 79.4        | 95.7         | 87.0        | <b>80.8</b> | 95.7        | <b>88.0</b> |

between rare words, while the missed NEG pairs in DiCo are between common words used in context.

### C. The negative class

Without the prior, the three reversible labels are rarely confused with each other; the errors that remain are almost all NEG against the rest. We tested six ways to improve that decision (Table V); the sixth, WordNet relation pairs, is discussed with the new backbones above.

Class weighting (weight 2 on NEG) and focal loss ( $\gamma = 2$ ) lowered HardCons on mDeBERTa by 3.9 and 1.1 points. Reverse-negative augmentation gained 0.8 F1 points on mDeBERTa but cost 4.7 SoftCons points, and hurt XLM-R-large on every metric. We then added 1,010 chunk pairs from the iSTS 2016 answers-students data, mapping EQUI to EQ, SPE1 and SPE2 to FE and BE, and SIMI, REL and OPPO to NEG. We did not use the headlines and images subsets, because they are the source of PhrasIS and contain its test pairs, and none of the converted pairs occurs in the DiCo training or development data. The extra data lowered F1 for both backbones, and a second mDeBERTa seed was also worse.

The last method was a stacked detector. A logistic regression and a gradient-boosted classifier predict NEG from the single-row log-probabilities of the seven ensemble members and from lexical features (token overlap, containment, length, number mismatch, negation words), with five-fold cross-validation grouped by `pair_id`. The best setting raised F1 by 0.6 points but lowered SoftCons by 3.9 and HardCons by 3.3. The ensemble’s own NEG score seems to carry the same lexical signal already.

Moving the NEG threshold  $\beta$  only trades one score for another (Fig. 5). Lowering it towards  $-1.5$  lifts HardCons to 90.3 and SoftCons to 99.3 while F1 falls to 76.9; raising it does the reverse. The mean of the three scores stays between 88 and 89 over the whole range.

### D. Error analysis

To see why, we pooled the clean-7 log-probabilities per source pair and inspected the 383 T1 development pairs. The errors are one-sided: 48 of the 106 NEG pairs are decoded as a reversible label (NEG recall 45%), against 13 of the 277 reversible pairs decoded as NEG. The pooled NEG score separates the two groups with an AUC of 0.90 on T1 and 0.87 on T3, and the best threshold on it gives a NEG F1 of only 74.6. This is why  $\beta$  cannot help: the problem is the ranking, not the cut-off. The missed NEG pairs are mostly iSTS SIMI and REL relations that look like entailment, such as *a creek/in a river*, *38/to 39* and *Australian senator/Australian lawmaker*; telling a hypernym from a sibling or a merely related phrase needs lexical knowledge that the phrases alone do not carry. Some labels also depend on the sentence the chunks came from, which DiCo does not provide: with all language views pooled, 27 of the 383 source pairs (7.0%) are missed by all eighteen multilingual clean runs, whatever the backbone or loss.

Among the reversible labels the picture is different. Given the prior, the ensemble picks the right one of EQ, FE and BE for 92.1% of T1 pairs and 88.8% of T3 pairs. Direction is essentially solved: only 2 of the 22 T1 errors confuse FE with BE; the rest confuse EQ with one of them, a question of granularity. T3 adds 13 BE-as-EQ errors, so Basque loses specificity. Finally, the ensemble members make similar mistakes: the error sets of mDeBERTa and XLM-R-large overlap with a Jaccard index of 0.61, and those of two mDeBERTa seeds with 0.69, which limits what further seeds of the same backbones can add.

### E. Hierarchical composition and ensemble selection

Table VI lists the systems we packaged as development submissions. The hierarchical composition helps without any selection. With the clean-7 ensemble as the gate and eleven

TABLE III  
DEVELOPMENT RESULTS WITH THE STRUCTURAL PRIOR, AND THE CODABENCH DEVELOPMENT LEADER ON 29 SEPTEMBER 2026. BEST VALUE PER COLUMN AMONG OUR SYSTEMS IN BOLD; COLUMNS WHERE ALL SYSTEMS TIE ARE NOT MARKED.

| System                              | T1 EN       |       |             | T2 ES       |       |             | T3 EU       |       |             | T4 Mixed    |       |             | ALL-4       |       |             |
|-------------------------------------|-------------|-------|-------------|-------------|-------|-------------|-------------|-------|-------------|-------------|-------|-------------|-------------|-------|-------------|
|                                     | F1          | S     | H           | F1          | S     | H           | F1          | S     | H           | F1          | S     | H           | F1          | S     | H           |
| mmBERT-base                         | 87.1        | 100.0 | 84.8        | 85.2        | 100.0 | 82.7        | 82.1        | 100.0 | 79.8        | 88.5        | 100.0 | 86.6        | 85.7        | 100.0 | 83.5        |
| mDeBERTa-v3-base NLI                | 92.3        | 100.0 | 91.0        | 90.5        | 100.0 | 88.8        | 89.6        | 100.0 | 87.7        | 90.9        | 100.0 | 89.2        | 90.8        | 100.0 | 89.2        |
| mDeBERTa, fine-grained val. (A)     | 91.7        | 100.0 | 90.3        | 89.6        | 100.0 | 87.7        | 89.9        | 100.0 | 88.1        | 90.3        | 100.0 | 88.4        | 90.4        | 100.0 | 88.6        |
| A + rev.-neg. aug.                  | 92.6        | 100.0 | 91.3        | 90.9        | 100.0 | 89.2        | 87.0        | 100.0 | 84.5        | 90.9        | 100.0 | 89.2        | 90.4        | 100.0 | 88.5        |
| A + NEG weight 2                    | 92.1        | 100.0 | 90.6        | 90.6        | 100.0 | 88.8        | 87.9        | 100.0 | 85.6        | 90.0        | 100.0 | 88.1        | 90.2        | 100.0 | 88.3        |
| A + focal ( $\gamma=2$ )            | 91.8        | 100.0 | 90.3        | 90.6        | 100.0 | 88.8        | 89.4        | 100.0 | 87.4        | 91.6        | 100.0 | 89.9        | 90.8        | 100.0 | 89.1        |
| A + iSTS answers-students           | 92.1        | 100.0 | 90.6        | 89.6        | 100.0 | 87.7        | 89.4        | 100.0 | 87.4        | 90.6        | 100.0 | 88.8        | 90.4        | 100.0 | 88.6        |
| A, symmetric arch.                  | 91.2        | 100.0 | 89.5        | 89.6        | 100.0 | 87.7        | 88.2        | 100.0 | 85.9        | 90.7        | 100.0 | 88.8        | 89.9        | 100.0 | 88.0        |
| A, entail2 head                     | 92.9        | 100.0 | 91.7        | 87.8        | 100.0 | 85.6        | 86.8        | 100.0 | 84.5        | 90.6        | 100.0 | 88.8        | 89.5        | 100.0 | 87.6        |
| A + WordNet pairs                   | 91.4        | 100.0 | 89.9        | 89.4        | 100.0 | 87.4        | 89.1        | 100.0 | 87.0        | 91.8        | 100.0 | 90.3        | 90.4        | 100.0 | 88.6        |
| XLM-R-large XNLI (B)                | 92.4        | 100.0 | 91.0        | 91.0        | 100.0 | 89.2        | 85.9        | 100.0 | 83.4        | 91.3        | 100.0 | 89.5        | 90.2        | 100.0 | 88.3        |
| B + rev.-neg. aug.                  | 91.0        | 100.0 | 89.2        | 88.3        | 100.0 | 85.9        | 85.3        | 100.0 | 82.3        | 86.6        | 100.0 | 83.8        | 87.8        | 100.0 | 85.3        |
| B + iSTS answers-students           | 90.6        | 100.0 | 88.8        | 90.5        | 100.0 | 88.8        | 89.0        | 100.0 | 87.0        | 91.8        | 100.0 | 90.3        | 90.5        | 100.0 | 88.7        |
| B + WordNet pairs                   | 91.0        | 100.0 | 89.2        | 89.5        | 100.0 | 87.4        | 86.1        | 100.0 | 83.0        | 90.8        | 100.0 | 88.8        | 89.3        | 100.0 | 87.1        |
| bge-reranker-v2-m3                  | 90.4        | 100.0 | 88.4        | 88.0        | 100.0 | 85.6        | 83.3        | 100.0 | 80.1        | 88.1        | 100.0 | 85.6        | 87.4        | 100.0 | 84.9        |
| Qwen3-Reranker-0.6B                 | 85.8        | 100.0 | 83.0        | 83.2        | 100.0 | 79.4        | 75.5        | 100.0 | 70.4        | 82.3        | 100.0 | 78.3        | 81.7        | 100.0 | 77.8        |
| Ens. clean-7 (A, B + 5 seeds)       | 93.3        | 100.0 | 92.1        | 91.2        | 100.0 | 89.5        | 90.7        | 100.0 | 88.8        | 92.5        | 100.0 | 91.0        | 91.9        | 100.0 | 90.3        |
| mDeBERTa, prior-aware               | 92.3        | 100.0 | 91.0        | 90.9        | 100.0 | 89.2        | 89.1        | 100.0 | 87.0        | 92.1        | 100.0 | 90.6        | 91.1        | 100.0 | 89.4        |
| prior-aware + WordNet pairs         | 91.2        | 100.0 | 89.5        | 90.3        | 100.0 | 88.4        | 87.9        | 100.0 | 85.6        | 91.9        | 100.0 | 90.3        | 90.3        | 100.0 | 88.4        |
| mDeBERTa, prior-aware, symmetric    | 93.0        | 100.0 | 91.7        | 91.8        | 100.0 | 90.3        | 90.1        | 100.0 | 88.1        | 92.5        | 100.0 | 91.0        | 91.9        | 100.0 | 90.3        |
| symmetric + WordNet pairs           | 92.9        | 100.0 | 91.7        | <b>92.4</b> | 100.0 | <b>91.0</b> | 89.6        | 100.0 | 87.7        | 91.5        | 100.0 | 89.9        | 91.6        | 100.0 | 90.1        |
| XLM-R-large, prior-aware            | 91.8        | 100.0 | 90.3        | 91.0        | 100.0 | 89.2        | 88.5        | 100.0 | 86.3        | 91.9        | 100.0 | 90.3        | 90.8        | 100.0 | 89.0        |
| XLM-R-large, prior-aware, symmetric | 92.4        | 100.0 | 91.0        | 91.2        | 100.0 | 89.5        | 87.6        | 100.0 | 85.2        | 91.0        | 100.0 | 89.2        | 90.5        | 100.0 | 88.7        |
| bge-reranker-v2-m3, prior-aware     | 91.1        | 100.0 | 89.5        | 89.6        | 100.0 | 87.7        | 86.2        | 100.0 | 83.8        | 91.8        | 100.0 | 90.3        | 89.7        | 100.0 | 87.8        |
| Qwen3-Reranker-0.6B, prior-aware    | 87.2        | 100.0 | 84.8        | 85.2        | 100.0 | 82.7        | 73.6        | 100.0 | 71.1        | 82.8        | 100.0 | 80.1        | 82.2        | 100.0 | 79.7        |
| Ens. prior-aware-5                  | 93.9        | 100.0 | 92.8        | 91.8        | 100.0 | 90.3        | 90.6        | 100.0 | 88.8        | 92.2        | 100.0 | 90.6        | 92.1        | 100.0 | 90.6        |
| Ens. prior-aware-5 + clean-7        | <b>94.2</b> | 100.0 | <b>93.1</b> | 92.1        | 100.0 | 90.6        | 90.4        | 100.0 | 88.4        | <b>92.8</b> | 100.0 | <b>91.3</b> | 92.4        | 100.0 | 90.9        |
| Ens. prior-14 (+ 2 symmetric)       | <b>94.2</b> | 100.0 | <b>93.1</b> | <b>92.4</b> | 100.0 | <b>91.0</b> | <b>91.3</b> | 100.0 | <b>89.5</b> | 92.5        | 100.0 | 91.0        | <b>92.6</b> | 100.0 | <b>91.2</b> |
| Leaderboard #1                      | 92.4        | 100   | 91.0        | 92.8        | 100   | 91.3        | 90.3        | 100   | 88.4        | 90.2        | 100   | 88.2        | 91.4        | 100   | 89.7        |

TABLE IV  
DECODING MODES FOR ONE DEBERTA-V3-BASE MODEL ON THE T1 DEVELOPMENT SET. BEST VALUE PER COLUMN IN BOLD.

| Decoding                | W-F1        | SoftCons     | HardCons    |
|-------------------------|-------------|--------------|-------------|
| independent             | 82.7        | 88.1         | 83.8        |
| twin                    | 82.1        | 93.5         | 86.3        |
| source                  | 82.1        | 93.5         | 86.3        |
| twin + structural prior | <b>92.4</b> | <b>100.0</b> | <b>91.0</b> |

TABLE V  
METHODS AIMED AT THE NEGATIVE CLASS: ALL-4 DEVELOPMENT SCORES WITHOUT THE PRIOR. BEST VALUE PER COLUMN IN BOLD.

| System                  | W-F1        | SoftCons    | HardCons    |
|-------------------------|-------------|-------------|-------------|
| A (mDeBERTa)            | 79.4        | 95.9        | 87.0        |
| + NEG weight 2          | 79.7        | 91.2        | 83.1        |
| + focal, $\gamma = 2$   | 78.2        | 94.4        | 85.9        |
| + reverse negatives     | 80.2        | 91.2        | 83.1        |
| + iSTS answers-students | 77.8        | 96.2        | 87.0        |
| + WordNet pairs         | 80.0        | 93.4        | 85.6        |
| B (XLM-R-large)         | 78.8        | 96.3        | 86.3        |
| + reverse negatives     | 74.3        | 92.7        | 79.6        |
| + iSTS answers-students | 77.6        | 92.8        | 84.3        |
| + WordNet pairs         | 77.3        | 95.8        | 84.7        |
| Clean-7 ensemble        | 80.8        | 95.7        | 88.0        |
| + stacked NEG detector  | <b>81.4</b> | 91.8        | 84.7        |
| with $\beta = -1.0$     | 79.1        | <b>98.5</b> | <b>89.8</b> |

TABLE VI  
DEVELOPMENT SUBMISSIONS, CONFIRMED WITH THE OFFICIAL SCORER (ALL-4, IN %). “SEL.” MARKS MEMBER LISTS CHOSEN ON THE DEVELOPMENT SET; “PRIOR” MARKS USE OF THE STRUCTURAL PRIOR. BEST VALUE PER COLUMN IN BOLD.

| Run id         | Sel. | Prior | W-F1        | SoftCons     | HardCons    |
|----------------|------|-------|-------------|--------------|-------------|
| clean7         |      |       | 80.8        | 95.7         | 88.0        |
| hier_fixed     |      |       | 81.4        | 96.8         | 89.5        |
| sel_clean      | ✓    |       | 82.0        | 97.7         | 90.2        |
| sel_hier       | ✓    |       | 83.7        | 98.6         | 91.7        |
| prior14        |      | ✓     | 92.6        | <b>100.0</b> | 91.2        |
| sel_prior      | ✓    | ✓     | 94.2        | <b>100.0</b> | <b>93.1</b> |
| sel_prior_v2   | ✓    | ✓     | <b>94.3</b> | <b>100.0</b> | <b>93.1</b> |
| Leaderboard #1 |      |       | 91.4        | 100.0        | 89.7        |

prior-aware models as the direction ensemble, it reaches 81.4 / 96.8 / 89.5, against 80.8 / 95.7 / 88.0 for clean-7 alone, with higher HardCons on all four tracks (T1 F1 is unchanged). Since both member lists are fixed in advance, this is the estimate we trust most for unseen data. Adding the WordNet and reranker models to the gate lowers it slightly (81.3 / 96.2 / 89.0).

Greedy selection per track adds more on the development set. Without the prior, selecting gate and direction members jointly gives 83.7 / 98.6 / 91.7, which is 2.9 F1 and 3.7 HardCons points above clean-7 and above the leaderboard leader on HardCons, though still 7.7 F1 points behind it. With



the prior, per-track selection reaches 94.3 / 100 / 93.1, ahead of the leader on every track. The selections pick odd members: mmBERT and the reverse-negative XLM-R model, both weak on their own, appear in some tracks, and the chosen sets change when more candidates are added. That instability is the usual sign of fitting the development set, so we expect selected systems to lose part of their margin on test data.

#### F. Cross-track pooling

Because a `pair_id` is shared by all tracks, decoding the four development files together pools up to eighteen views of each source pair. In a four-model ensemble with the prior this added about one point of both F1 and HardCons over per-track decoding (92.7 / 100 / 91.3 against 91.8 / 100 / 90.3). It also uses the other tracks’ inputs to predict a given track and makes the four tracks’ predictions identical, so we leave it out of the reported results until the organizers say whether it is allowed.

### VII. DISCUSSION

Two caveats apply to every number above. First, all model, ensemble and threshold choices were made on the development set, which has 277 reversible pairs per track; the standard error of HardCons at that size is about 2.6 points, so differences of one or two points between single runs are within noise, and the reported scores are optimistic. Grouped cross-validation over the training and development data would give firmer estimates. Second, the lead over the leaderboard depends entirely on the structural prior. Without it, our best selected system is 7.7 points of F1 behind and our best fixed system 10.0 points, and the gap is the NEG decision. Selected ensembles are also the least trustworthy numbers in this report: their member lists change when candidates are added, and several members are weak on their own. For the evaluation phase we would submit fixed ensembles, such as `hier_fixed`, trained on training and development data together.

The negative results are consistent with what is known about NLI models on short, lexically overlapping inputs [11], [12]: the hard NEG cases are phrases that are related, similar or opposite, and more weight, more data of the same kind, WordNet relations between rare words or extra surface features did not teach the models to separate them. Two further directions stay within our model budget: lexical-relation pairs drawn from the vocabulary of DiCo itself rather than the whole wordnet, and grouped cross-validation over training and development data to select ensembles on held-out folds instead of on the development set. Methods that change the model’s inductive bias rather than its training signal, such as larger instruction-tuned models used as teachers, are the obvious next step.

### VIII. CONCLUSION

Consistency in DiCo-NLI can be guaranteed by decoding the two directions of a pair together, which raises SoftCons and HardCons at no training cost. With that in place, the task

reduces to deciding whether a phrase pair stands in an entailment relation at all, and the evidence we collected says this decision is limited by lexical knowledge about common words in context. Splitting the work between a negative-class gate and a direction model trained only on the entailment labels is the one change that helped without selection, lifting the fixed clean ensemble to 81.4 / 96.8 / 89.5 on the development set. Selected ensembles reach 83.7 / 98.6 / 91.7 without and 94.3 / 100 / 93.1 with the layout-based prior. Whether that prior may be used in the evaluation phase is a question for the organizers.

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